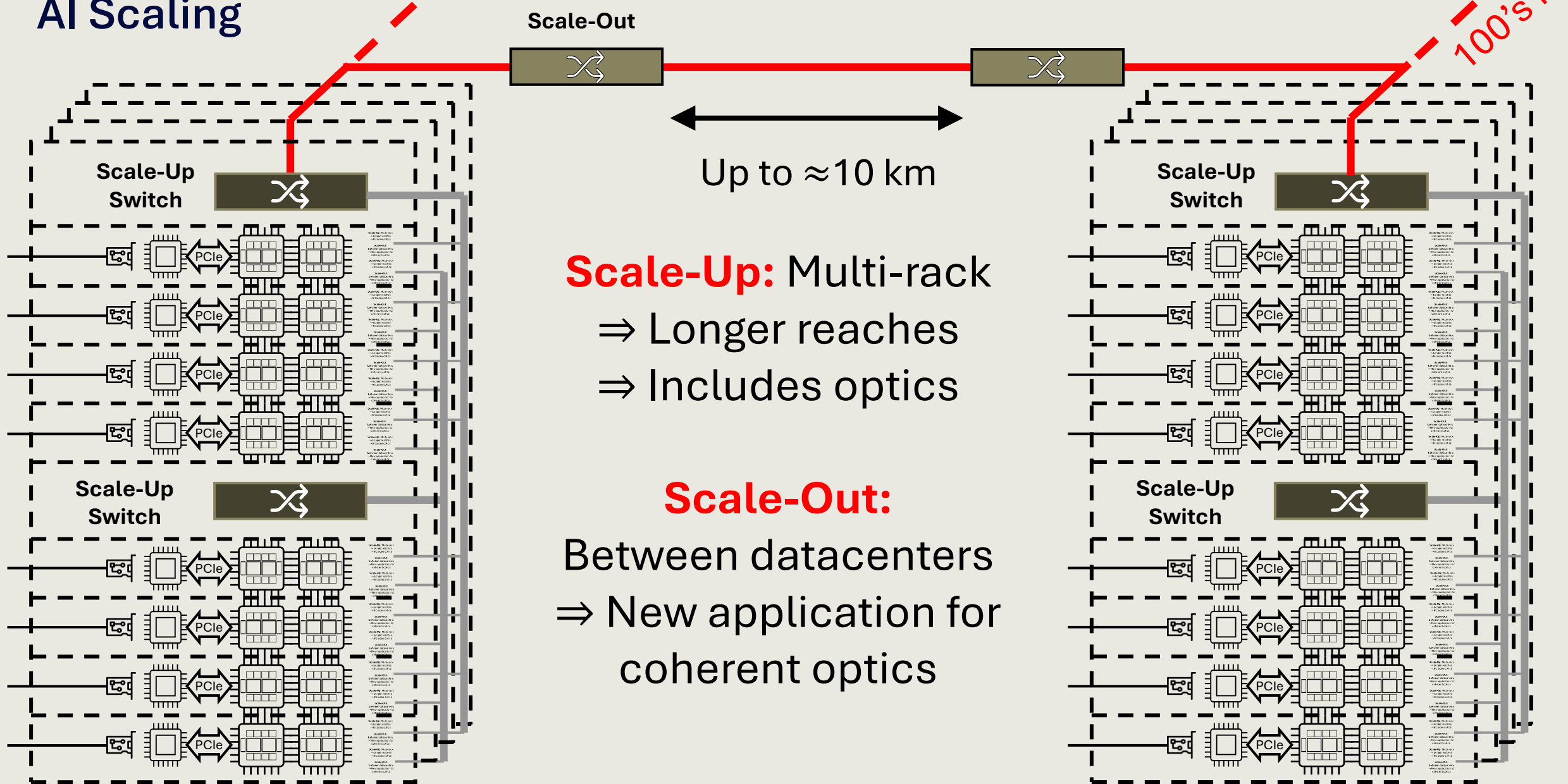


Scaling AI with CPO

David Kulansky

Director of Product Management, Qualcomm

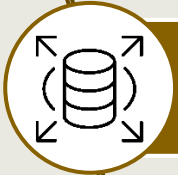
AI Scaling



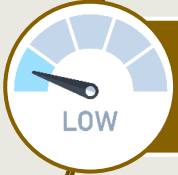
Connectivity Demands for AI



Higher per-lane data rates
⇒ harder channels



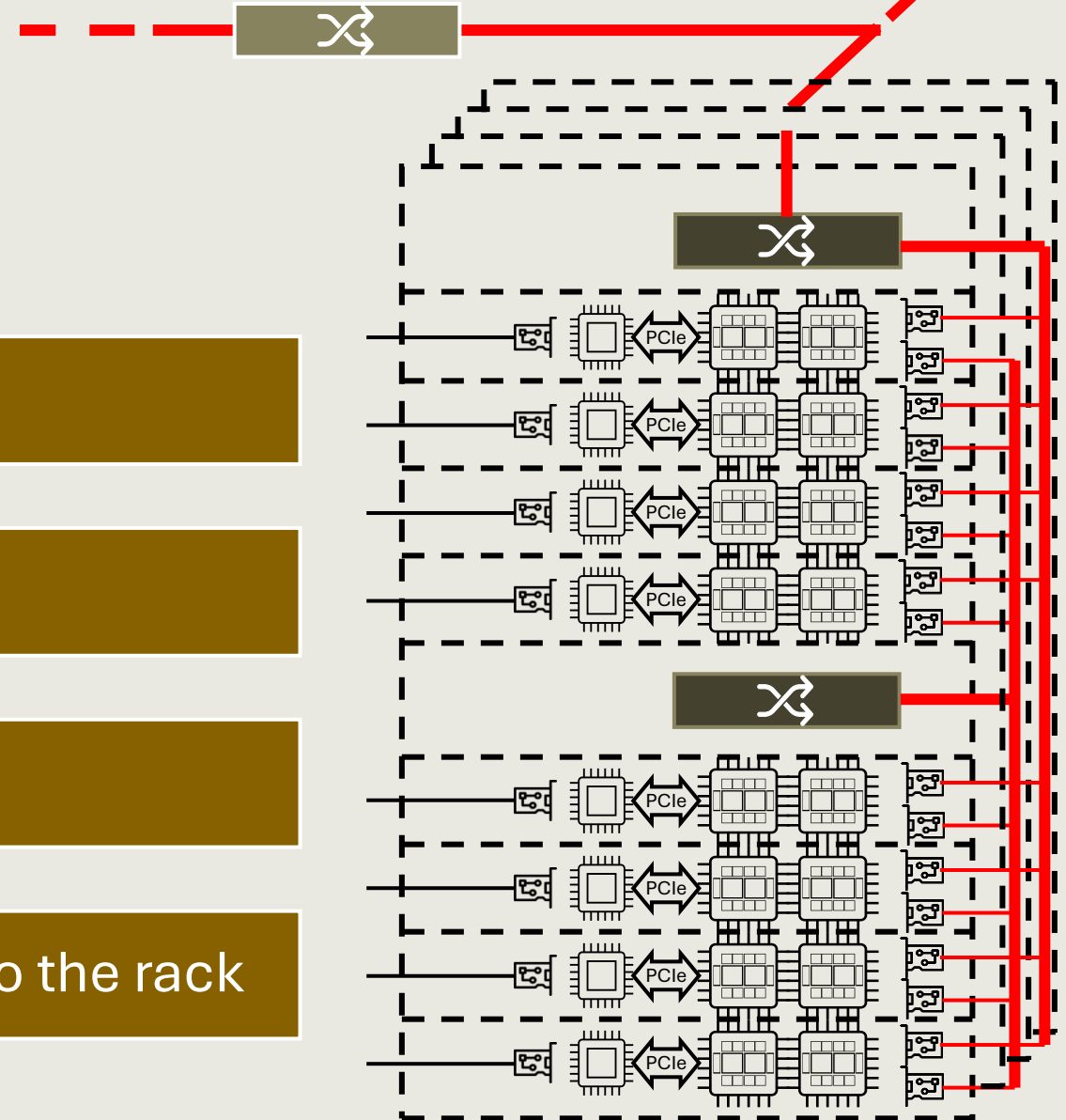
Scale up and out to larger clusters
⇒ harder channels



Low latency



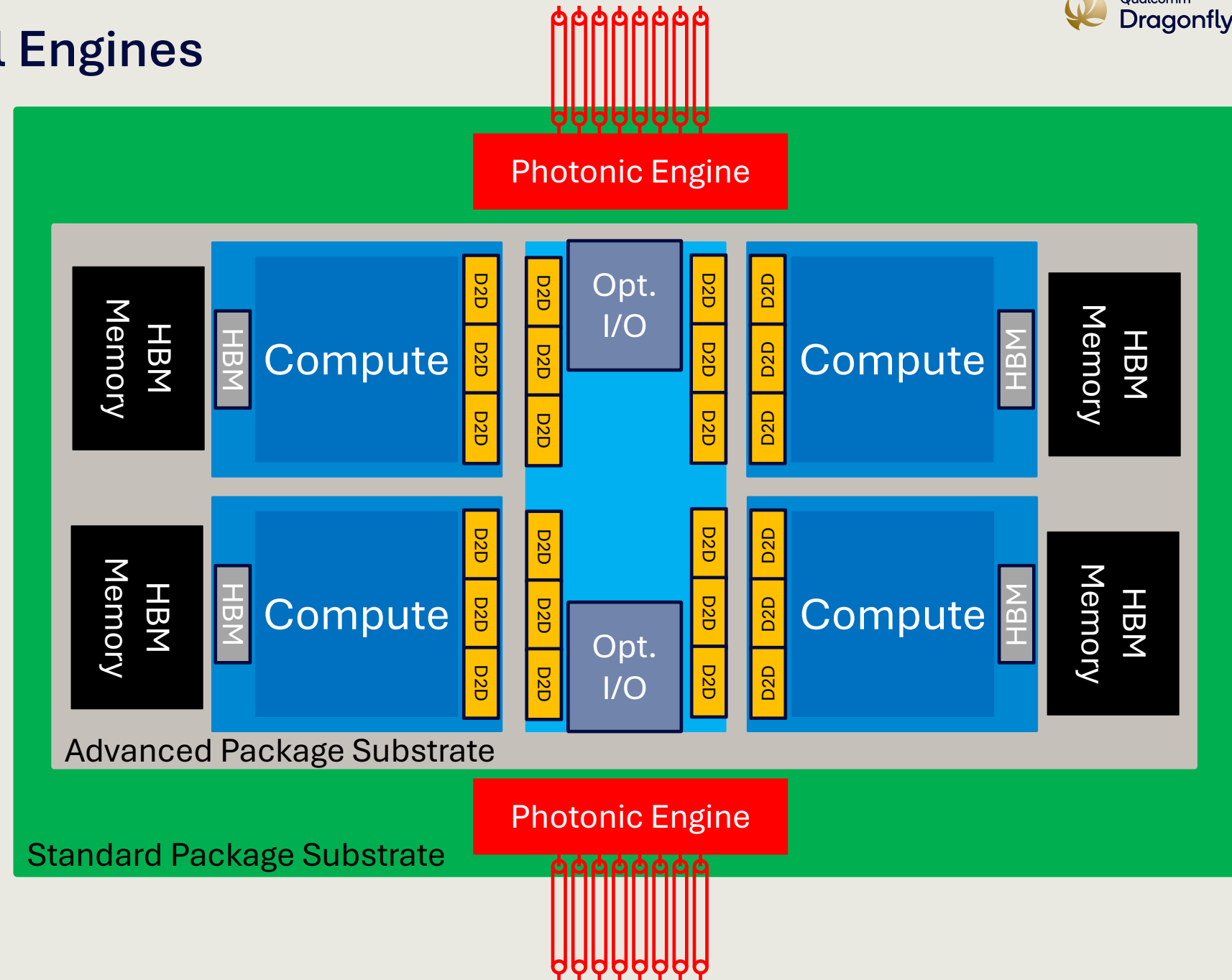
Penetration of optical connectivity into the rack



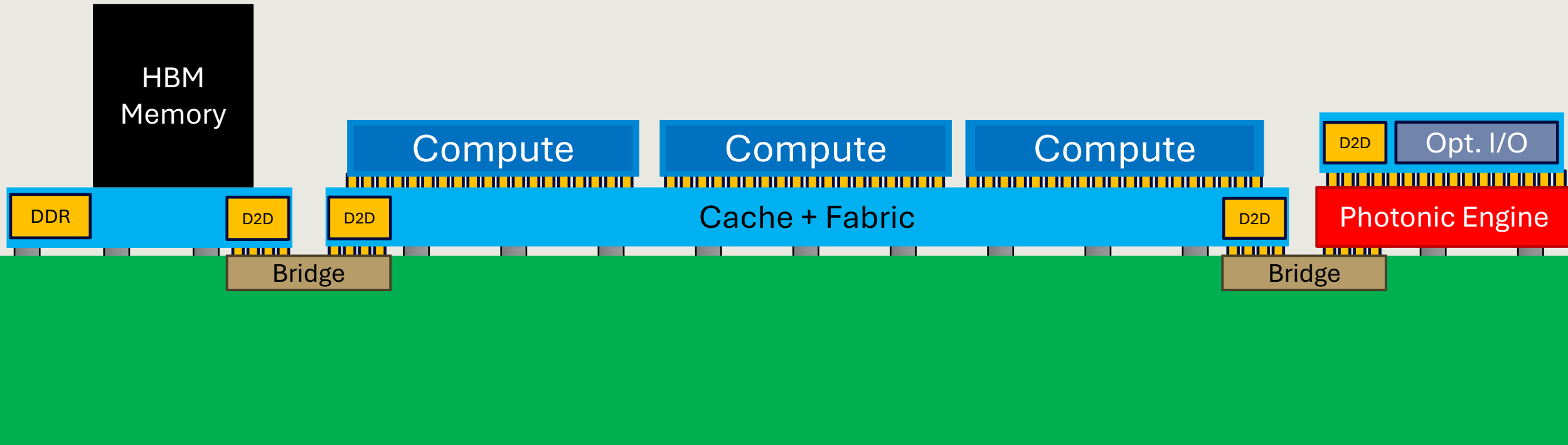
Integration of Optical Engines

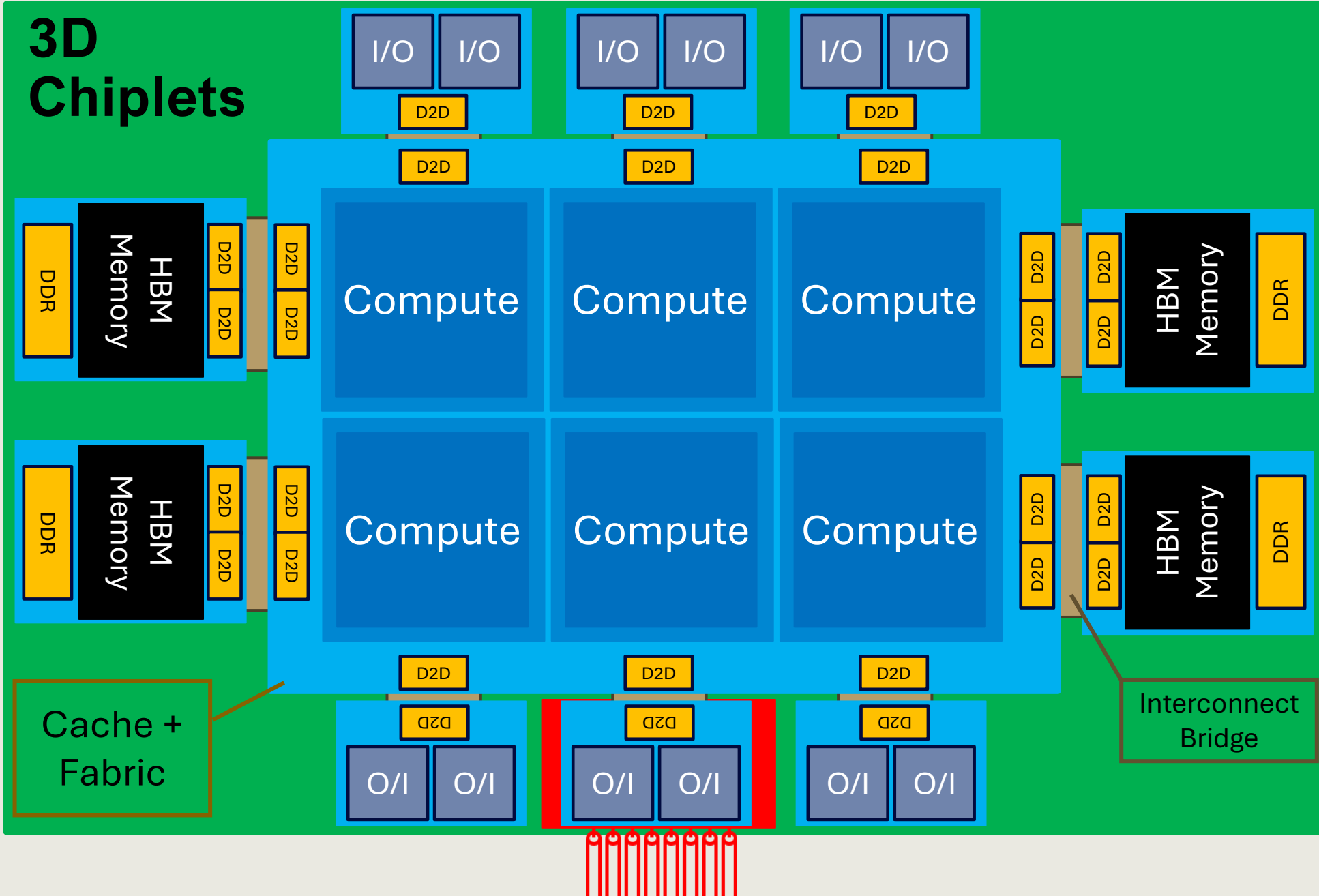
Leverage:

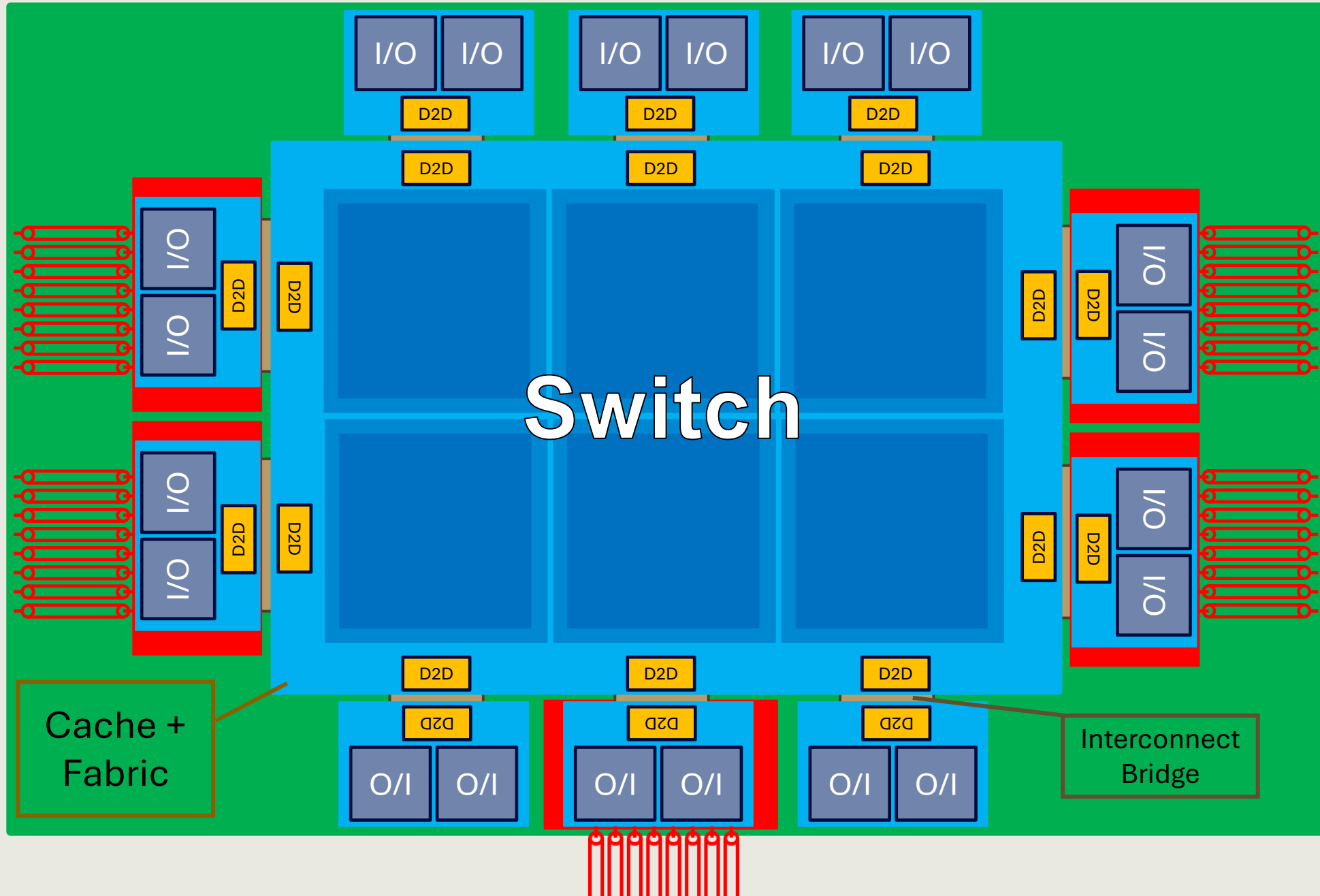
- Chiplet architectures
- Heterogeneous integration
- Optical SerDes technology
- Photonics



3D Chiptlet Architectures

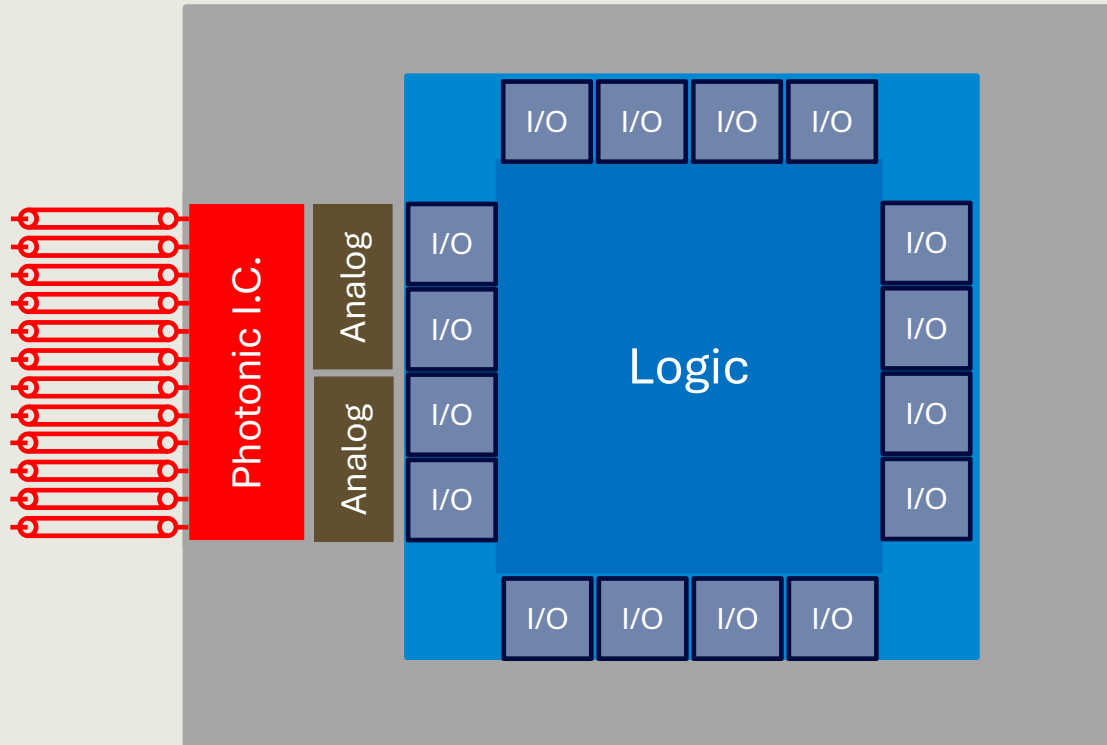




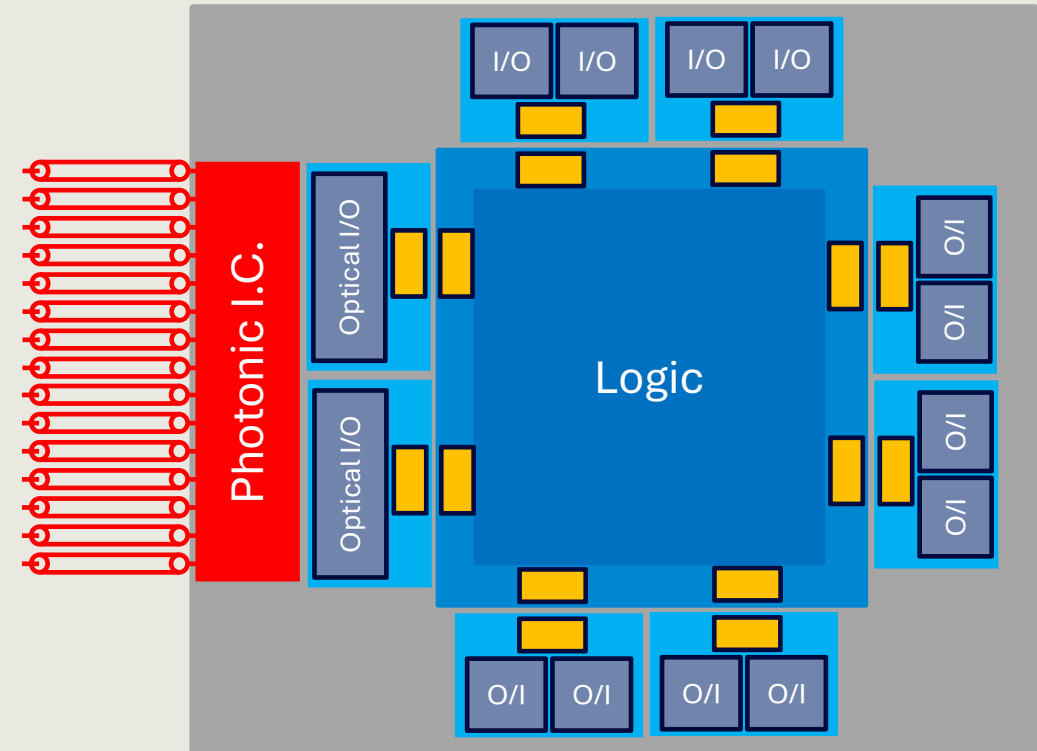


Co-Packaged Optics (CPO) Chiplet Architectures

Direct-Drive CPO

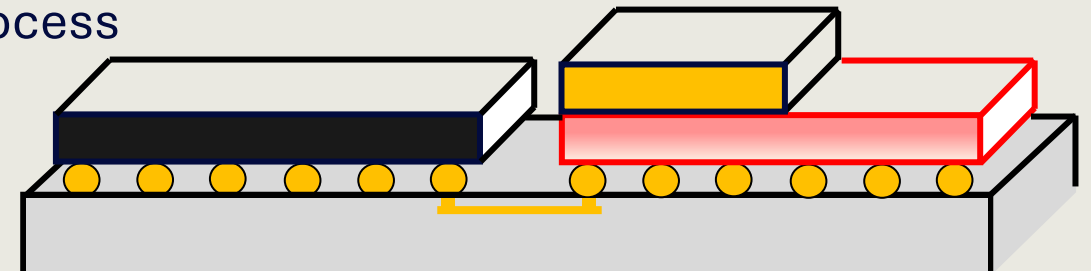
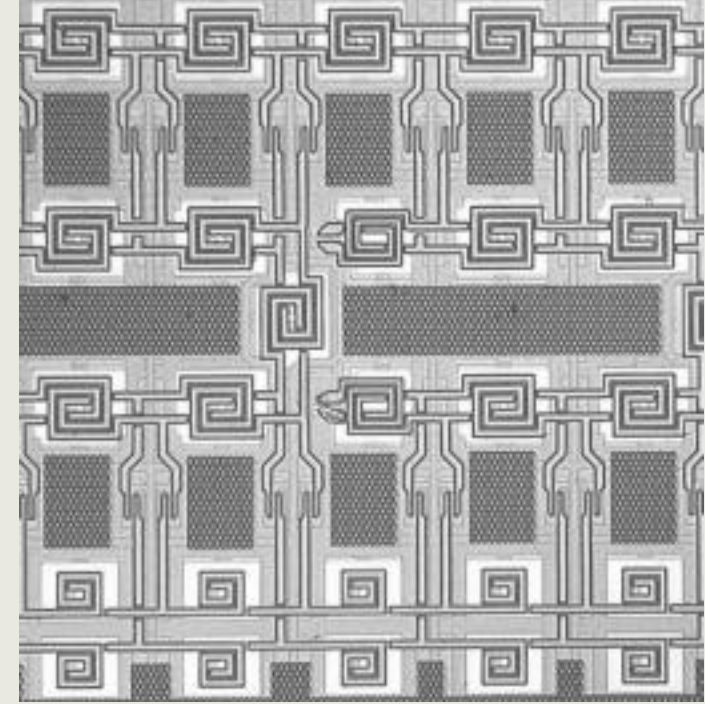


Digital-Drive CPO



Chiplet Implementation of High-Speed I/O Circuits

- Pervasive use of inductors
 - Need for accurate electromagnetic simulation of all interconnect parasitics, including packaging
 - Dense layout difficult due to potential for inductive coupling on-die or on-package
- Dedicated higher supply voltages desirable for additional SNR, lower jitter
- Heterogeneous integration opens the door to use of $(N - 1)$ CMOS, SiGe or SiPho technologies
- Preference for SerDes pre-validated in a particular process

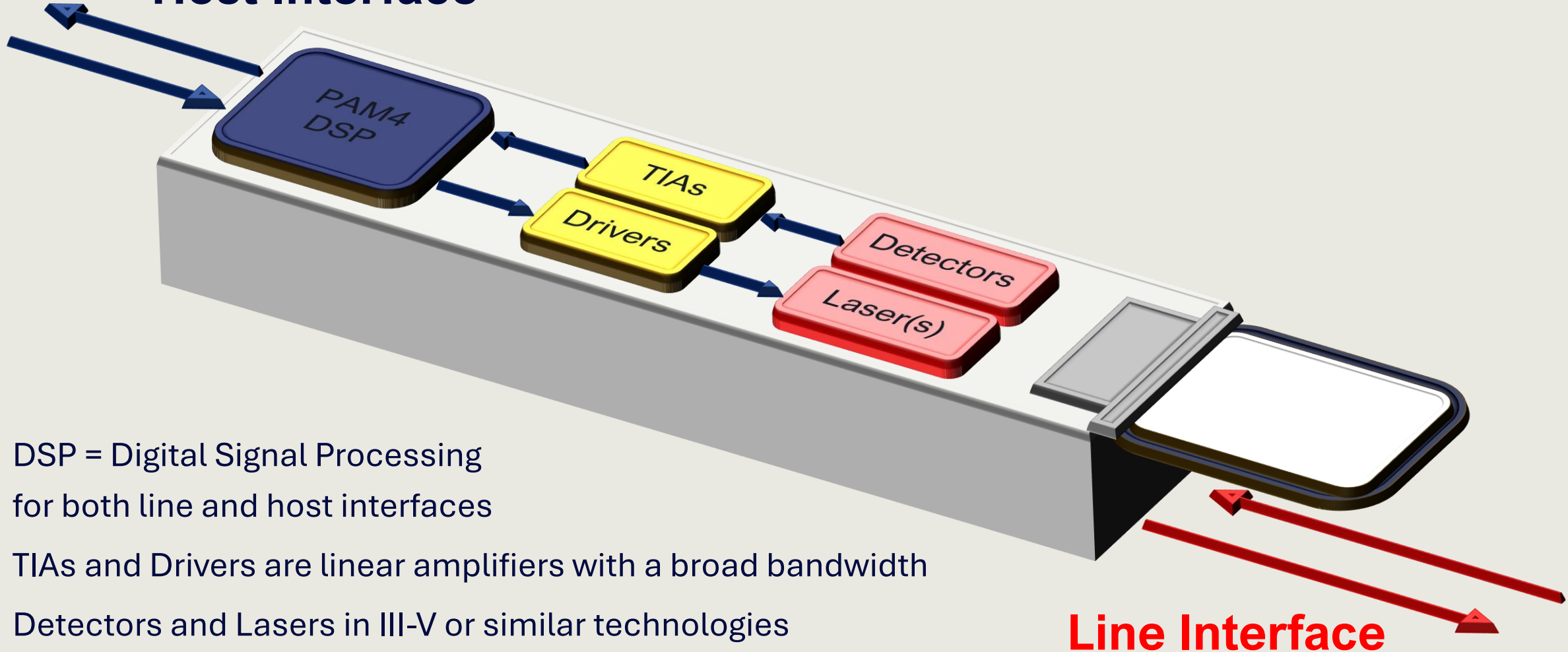




Chiplet Architectures for Optical Transceivers

Traditional Optical Module Anatomy

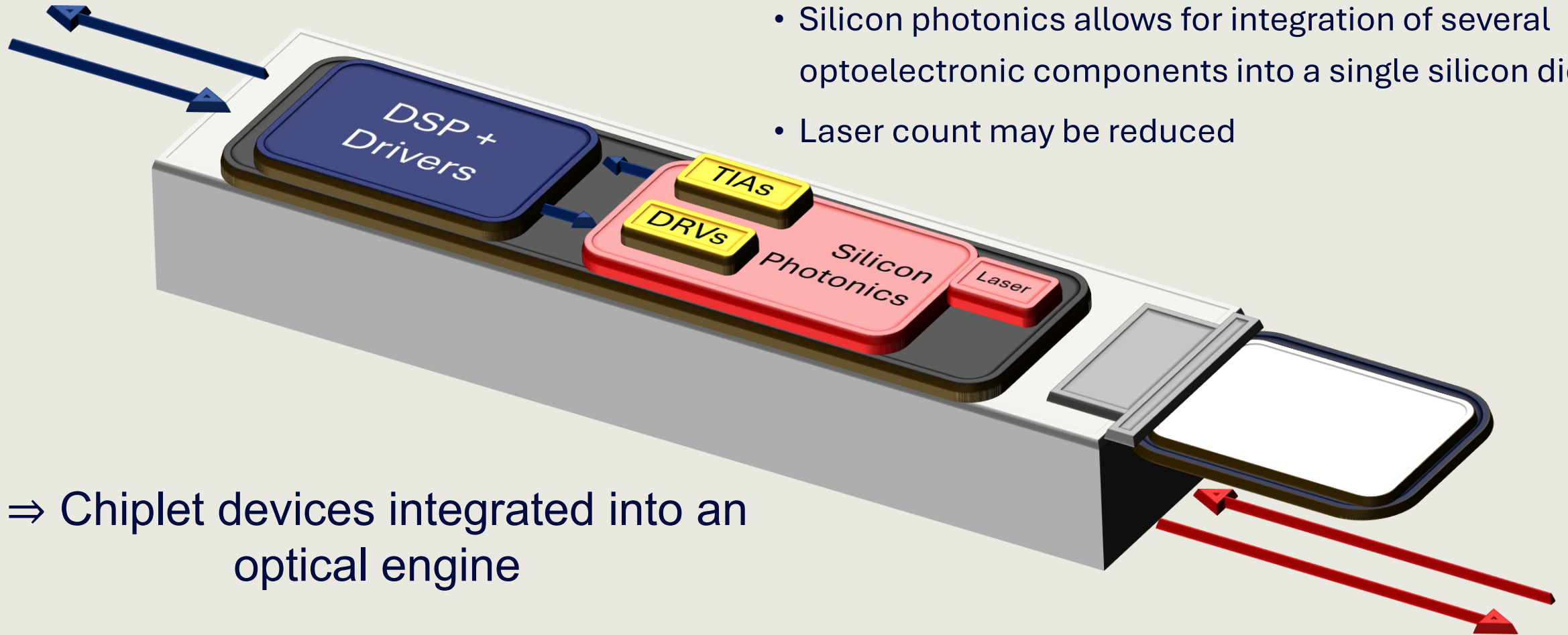
Host Interface



- DSP = Digital Signal Processing for both line and host interfaces
- TIAs and Drivers are linear amplifiers with a broad bandwidth
- Detectors and Lasers in III-V or similar technologies

Line Interface

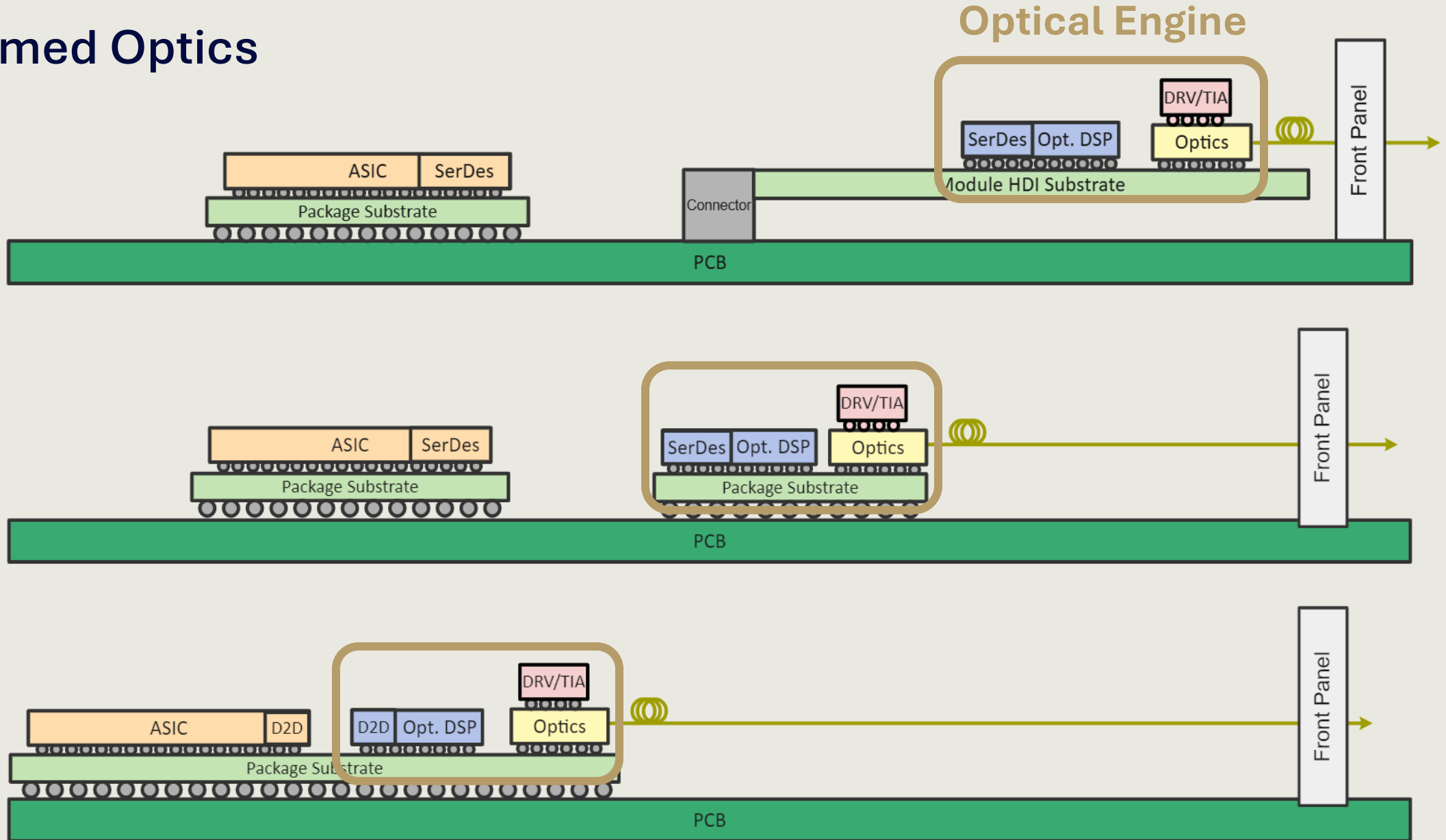
Optical Engine in a Module



- DSP = Digital Signal ProcDSP with integrated drivers reduces the component count
- Silicon photonics allows for integration of several optoelectronic components into a single silicon die
- Laser count may be reduced

⇒ Chiplet devices integrated into an optical engine

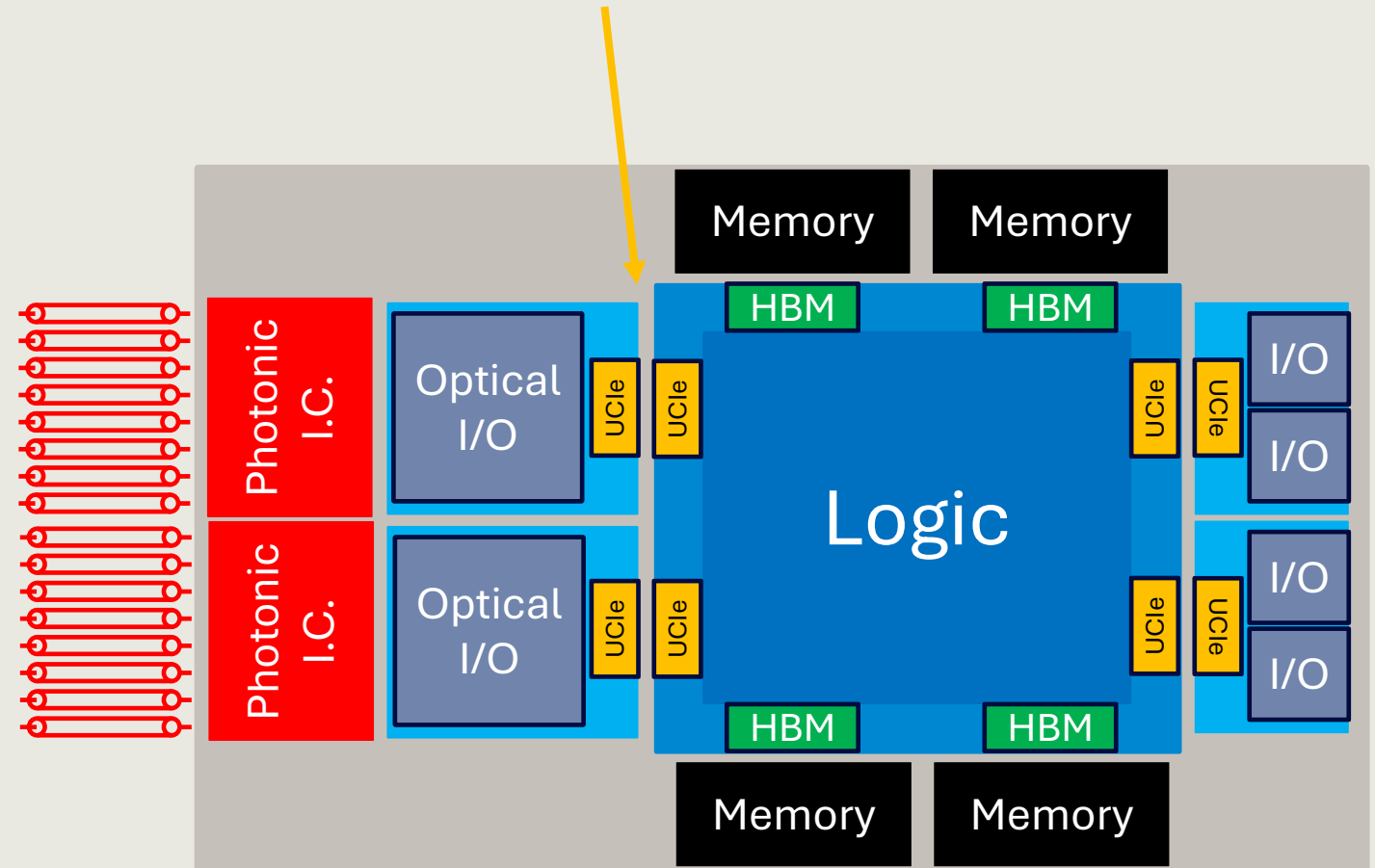
Retimed Optics



CPO Chiplets

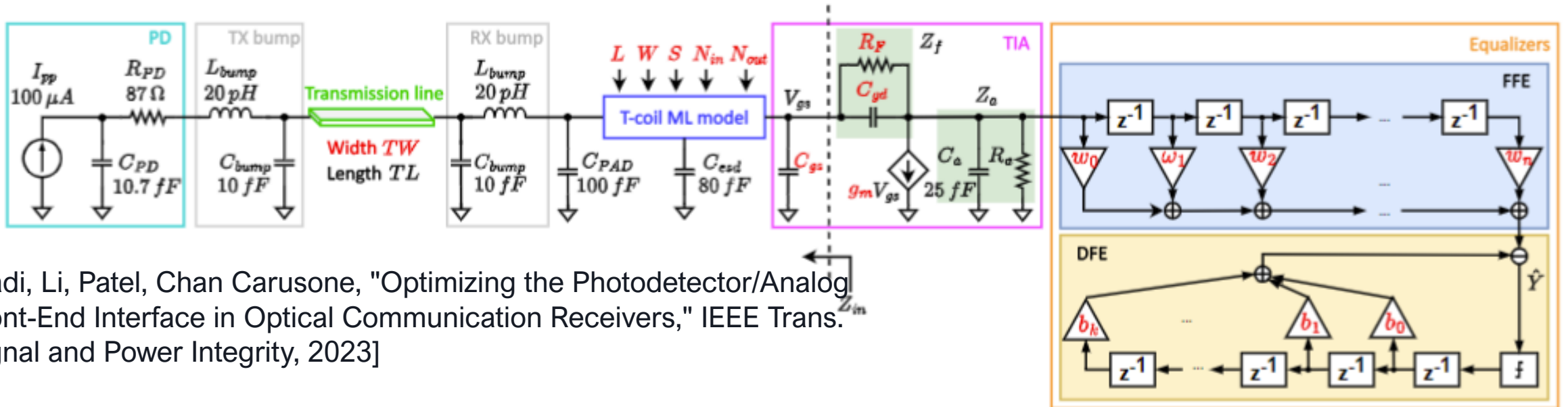
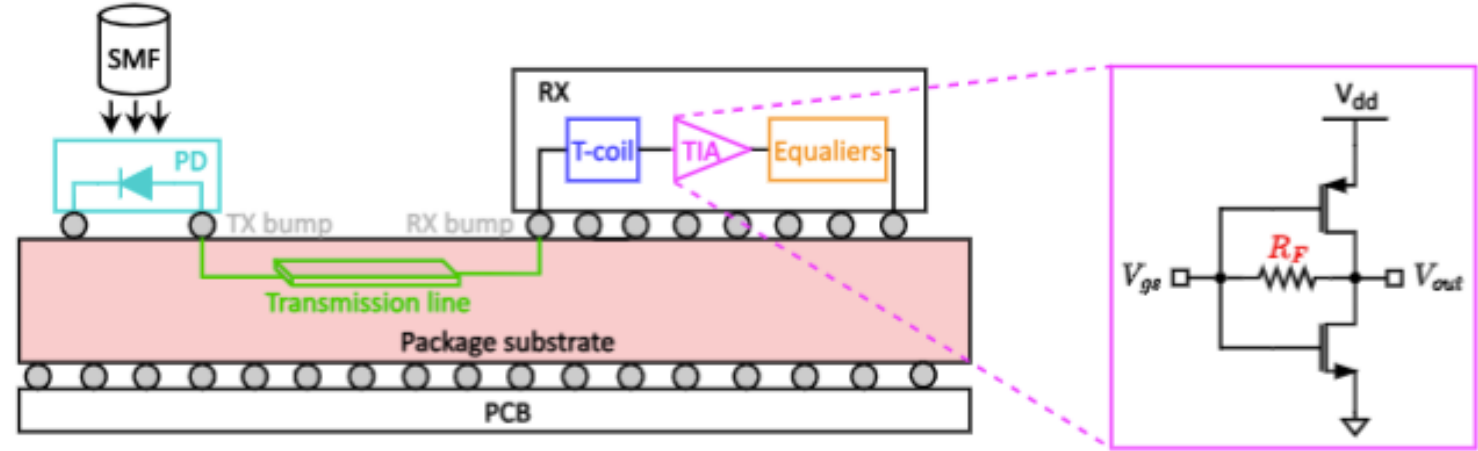
- Mix-and-match with electrical I/O chiplets for different applications
- Optical I/O density can be increased with new solutions:
 - Dense fiber arrays
 - Multiple wavelengths
 - Higher optical signaling rates

Bandwidth (“Beachfront”)
Density 20 Tbps/mm



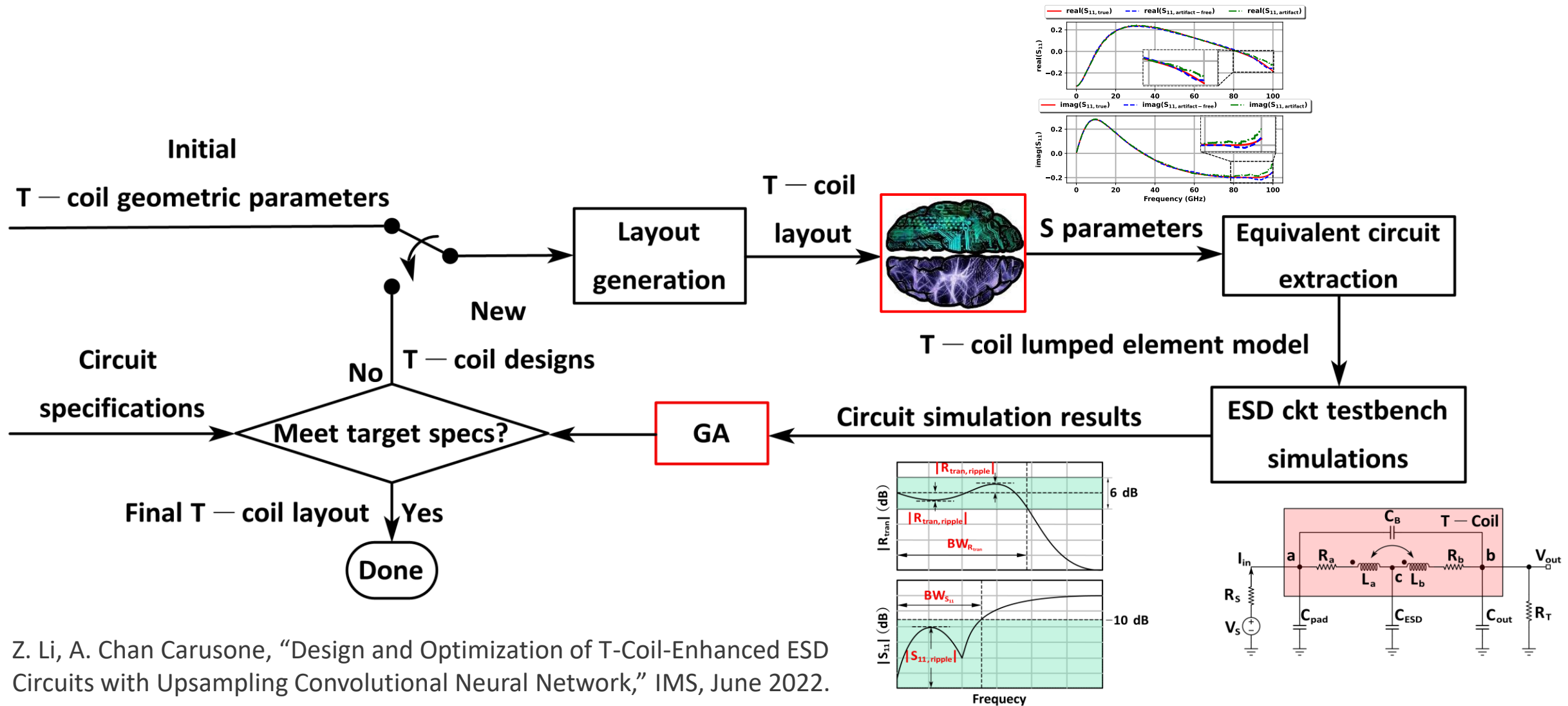
Complete Optical Receiver Co-Design

- Dozens of design parameters
- Co-optimization to minimize output noise



[Radi, Li, Patel, Chan Carusone, "Optimizing the Photodetector/Analog Front-End Interface in Optical Communication Receivers," IEEE Trans. Signal and Power Integrity, 2023]

T-Coil Optimization Flow Chart

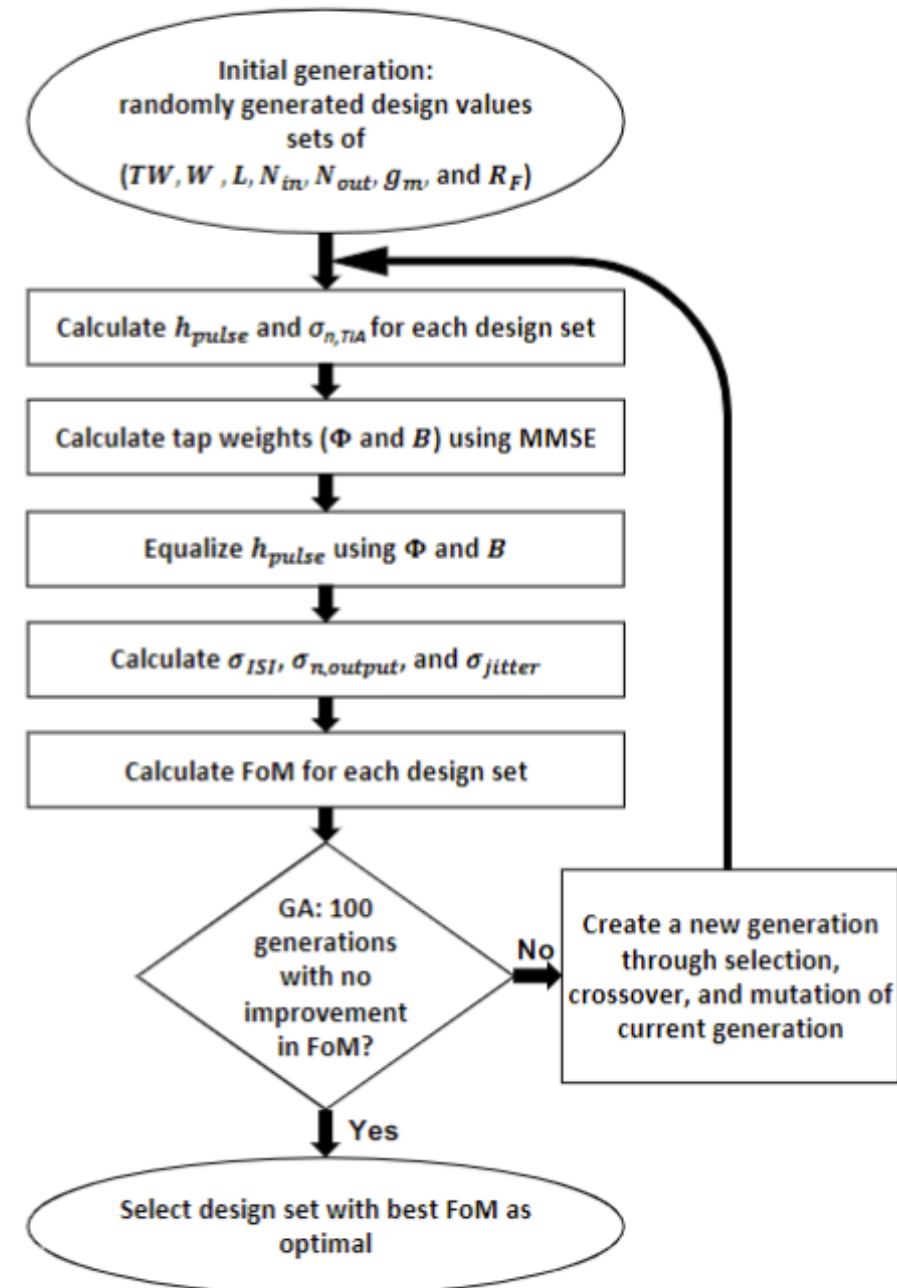
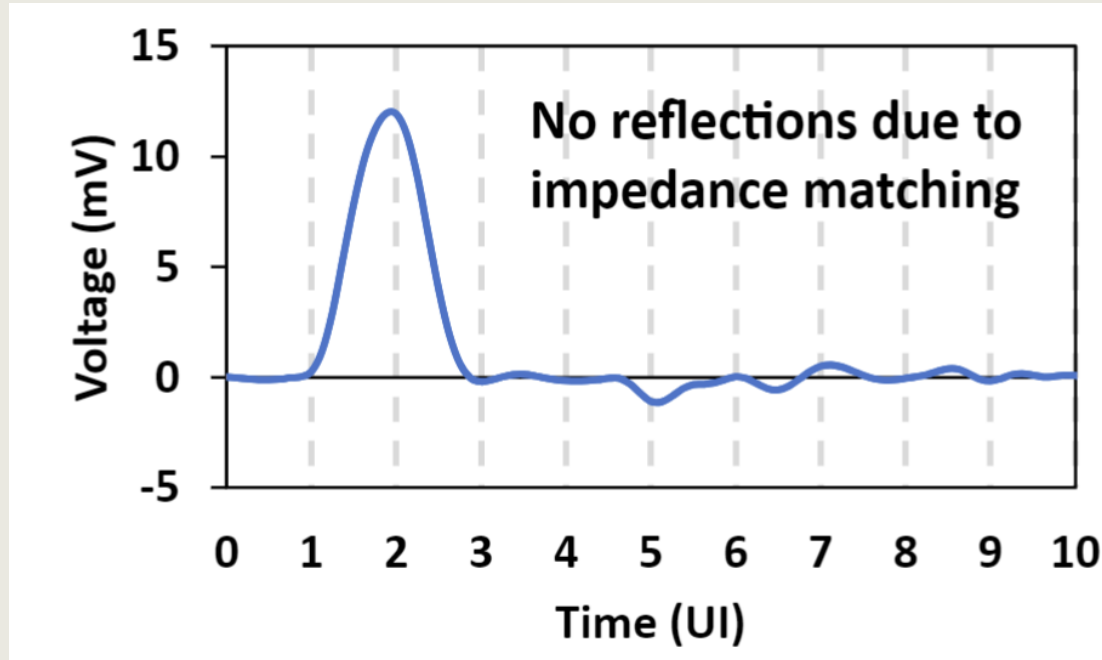


Z. Li, A. Chan Carusone, "Design and Optimization of T-Coil-Enhanced ESD Circuits with Upsampling Convolutional Neural Network," IMS, June 2022.

Optimization Flow

[B. Radi, Z. Li, D. Patel and A. Chan Carusone, "Optimizing the Photodetector/Analog Front-End Interface in Optical Communication Receivers," in *IEEE Transactions on Signal and Power Integrity*, Aug. 2023]

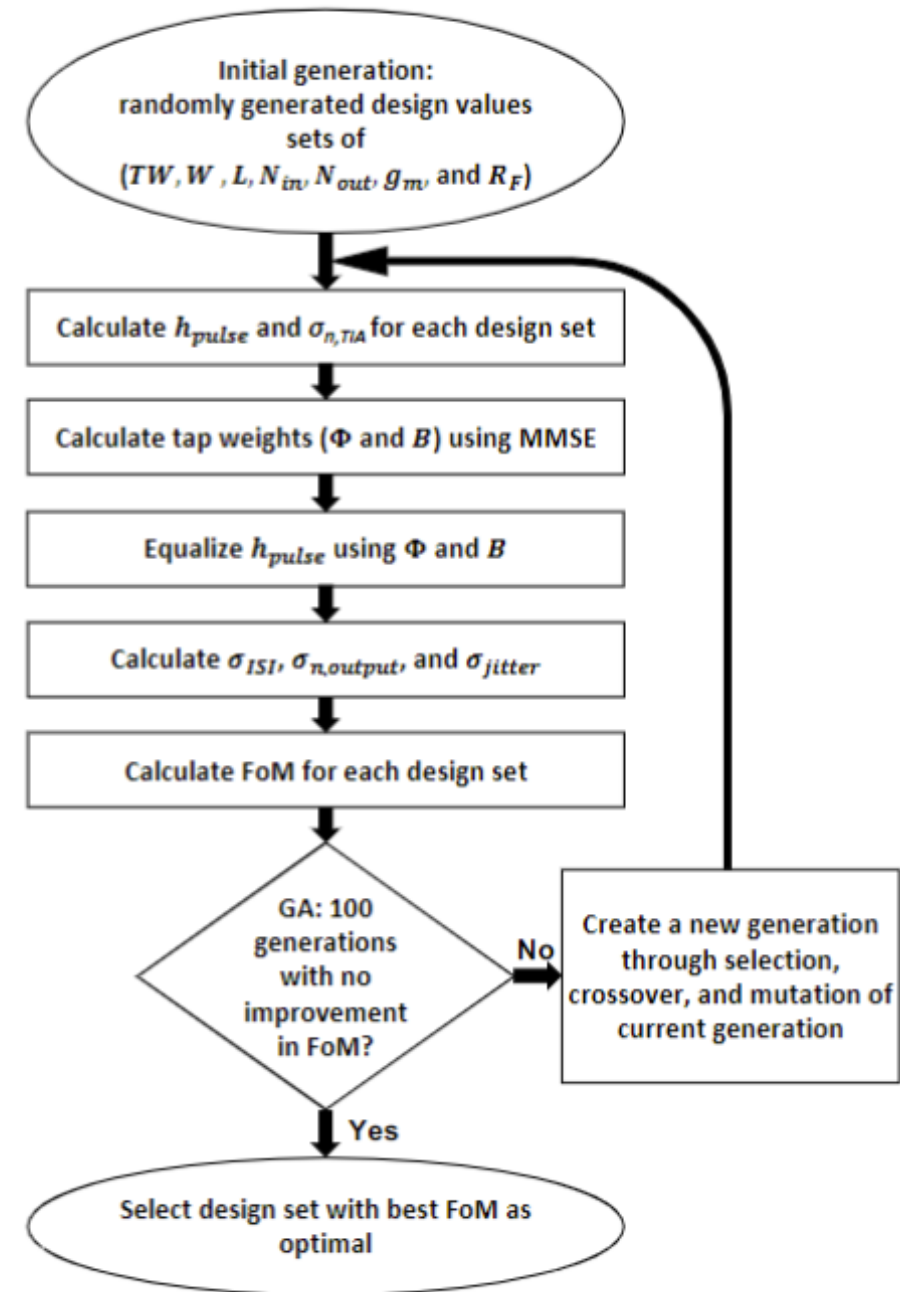
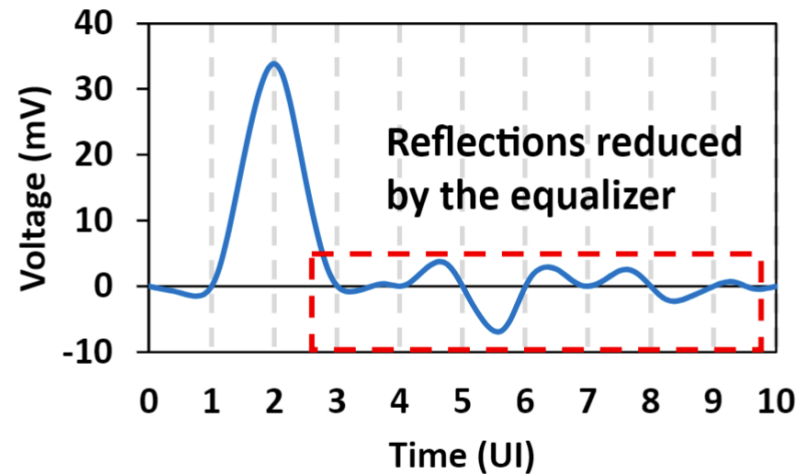
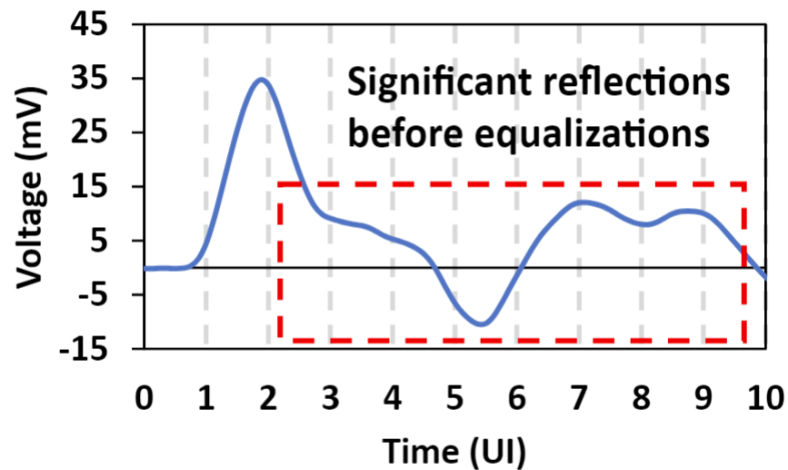
- Different optima depending on the DSP
- No DSP requires impedance-matched trace



Optimization Flow

[B. Radi, Z. Li, D. Patel and A. Chan Carusone, "Optimizing the Photodetector/Analog Front-End Interface in Optical Communication Receivers," in *IEEE Transactions on Signal and Power Integrity*, Aug. 2023]

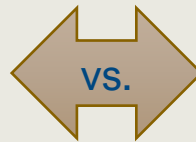
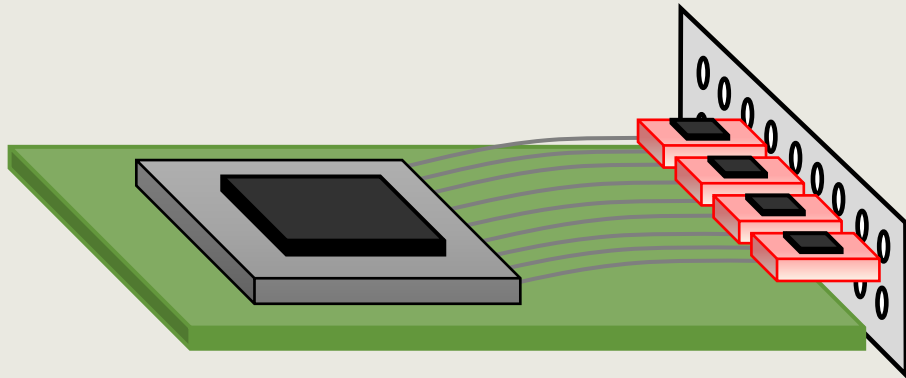
- Different optima depending on the DSP
- With DSP equalization, better to leave some reflections:



Opportunities and Challenges for CPO

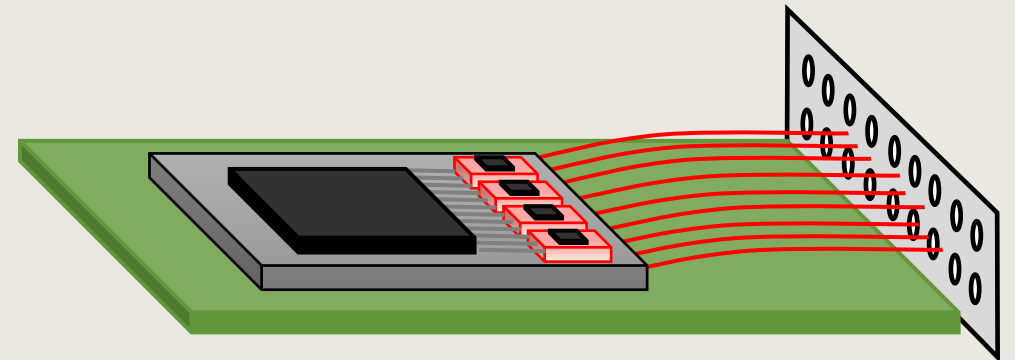
Pros

- May eliminate retimers
 - Lower system power
 - Lower latency
 - Potential to lower cost
- Potential to improve aggregate I/O bandwidth
 - Alleviates faceplate density issues



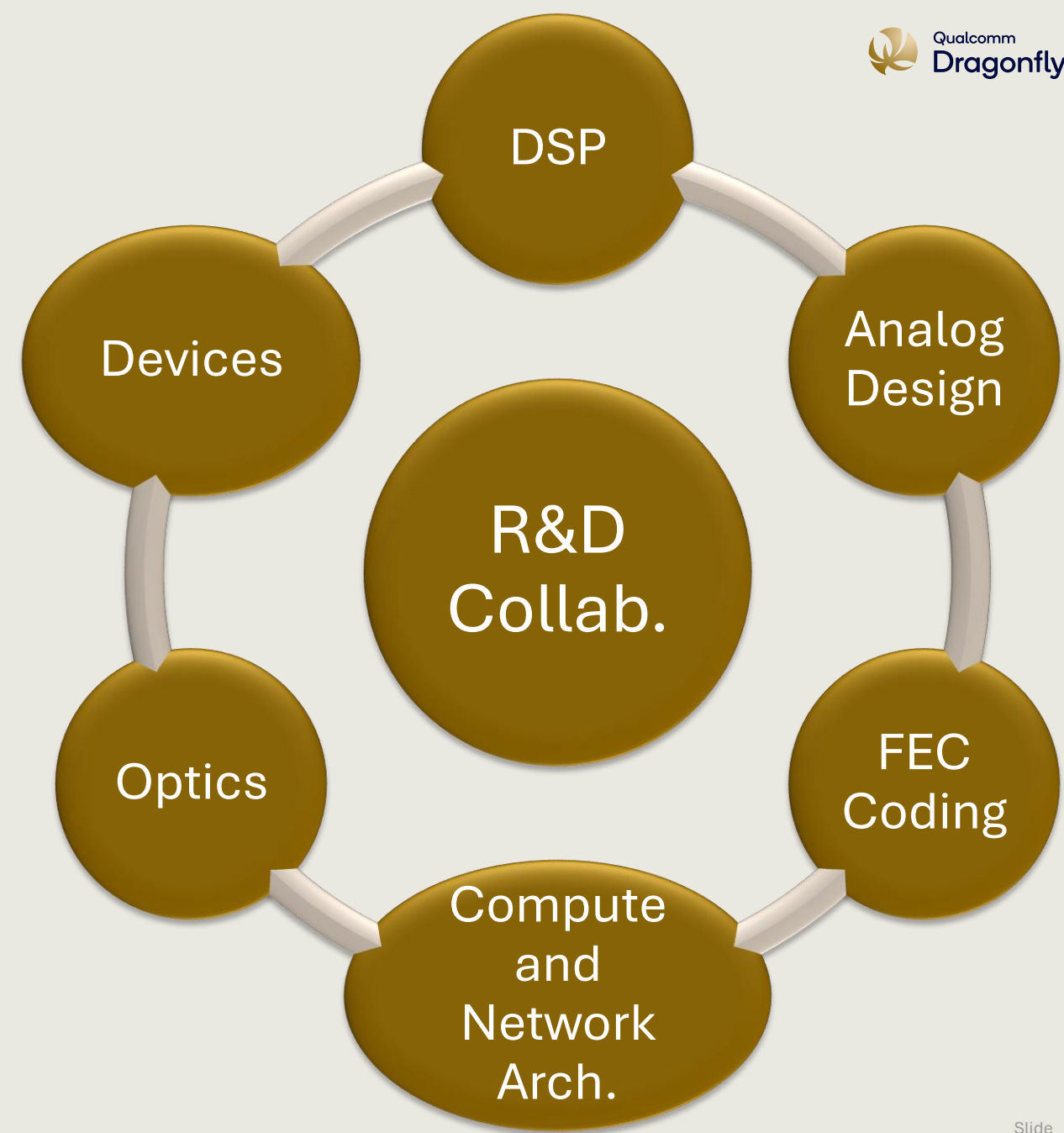
Cons

- Concentrates power/heat within the ASIC package
- Switch assembly more complex
- Field service more difficult
- Reduced ecosystem for innovation



Summary

- Megatrends driving connectivity:
 - Massive scale-out of compute for AI
 - AI workloads demanding low latency, low power, high throughput
 - Disaggregated storage is another driver
- Proliferation of new connectivity technologies
 - Electrical links can work with 45dB loss
 - Optical connectivity needed for shorter reaches as rates climb
 - Highly-integrated optical engines
 - Direct-drive for lowest power & latency





Thank you

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